Notes: Alessandria and Choi (QJE, 2007)

Do Sunk Costs of Exporting Matter for Net Export Dynamics?

Contents

1 Big picture

- **Question.** Do sunk costs of entering export markets materially change the aggregate dynamics of net exports and real exchange rates?
- **Motivation.** A classic exporter-hysteresis view argues that firms do not enter and exit foreign markets instantly. Because entry is costly and exit is delayed, net exports should respond slowly to exchange-rate movements.
- **Core setting.** The paper embeds plant-level export participation into a two-country international business cycle model. Firms are heterogeneous, face idiosyncratic productivity shocks, and choose whether to export.
- **Key friction.** A nonexporter must pay a large sunk entry cost τ_0 to enter a foreign market. An incumbent exporter pays a smaller continuation cost $\tau_1 < \tau_0$ to remain in that market.
- **Micro result.** The model generates persistent exporter status, realistic exporter premia, and business-cycle variation in entry and exit. Exporters are larger and more productive than nonexporters, and export participation is sticky.
- **Main macro result.** Despite realistic exporter dynamics, the aggregate paths of net exports and the real exchange rate are almost unchanged relative to a standard no-export-cost international business cycle model.
- **Why the macro effect is small.** In the benchmark calibration, goods produced within the same country are close substitutes, and home and foreign aggregates are relatively poor substitutes. As a result, the economy can satisfy import demand using many goods from few suppliers or few goods from many suppliers.
- **Important exception.** Export participation matters more when the number of varieties itself has a strong utility or disutility effect. Thus, the key parameter is not only the sunk cost, but also how consumers value extra foreign varieties.
- **Data comparison.** The model also produces exporter-count dynamics that are broadly consistent with U.S. exporters to major OECD destinations from 1995 to 2003.
- **Economic takeaway.** Firm-level export hysteresis is real, but it does not automatically create large aggregate trade-balance hysteresis. The paper shifts attention from participation costs alone to lags in expanding trade flows and to the welfare/measurement role of product variety.

2 Incremental contribution of the paper

- **From partial equilibrium hysteresis to general equilibrium.** Earlier models by Baldwin, Krugman, and Dixit studied export entry and exit in partial equilibrium. This paper asks whether the same mechanism survives quantitatively inside a full international business cycle model.
- **A bridge between micro exporter facts and macro trade dynamics.** The model is disciplined by micro facts: few firms export, exporters are larger and more productive, and exporter status is persistent.

- **A clear negative result.** The paper shows that adding sunk export costs does not noticeably change aggregate net export and real exchange rate dynamics under standard international business cycle calibrations.
- **A decomposition of the source of aggregate effects.** The authors separate the role of exporter participation from the role of variety. They show that the taste for variety, rather than sunk export costs by itself, is the main channel through which changing exporters can affect aggregates.
- **A benchmark against fixed-exit alternatives.** The paper compares the endogenous sunk-cost model to a no-cost model and to a fixed-exit model. This isolates whether endogenous selection of exiting exporters matters for macro dynamics.
- **A measurement contribution.** The paper discusses how standard price indices may miss changes in the number of varieties and evaluates whether this affects real exchange rate dynamics.
- **New exporter-cycle evidence.** The paper compares model-generated exporter counts with U.S. exporter data by destination market, showing that exporter participation moves with trade, source-country GDP, and the real exchange rate in ways that the model partly captures.
- **Policy/theory implication.** The findings caution against inferring large aggregate effects from micro frictions. To understand trade-balance dynamics, researchers may need to model delivery lags, customer accumulation, pricing-to-market, or other frictions beyond the fixed cost of entering an export market.

3 Data and measurement

3.1 Exporter facts used to discipline the model

- Most firms do not export. The paper cites evidence that only about 21% of U.S. manufacturing plants exported in the 1992 Census of Manufactures.
- Export status is persistent. In a balanced panel from the Annual Survey of Manufactures, about 87.4% of exporters continue exporting the next year and about 86.1% of nonexporters remain nonexporters.
- Exporters are larger and more productive than nonexporters. They have higher productivity, more workers, more capital per worker, and more output.
- The model is calibrated to match the export share, exporter entry and exit rates, and exporter premia.

Interpretation: The paper does not deny exporter hysteresis. It builds a model that intentionally generates it, then asks whether the aggregate trade implications are quantitatively large.

3.2 Aggregate macro data

- Aggregate business-cycle moments are computed for the United States from 1975:1 through 2004:3.
- The variables include output, consumption, investment, labor, net exports relative to GDP, the terms of trade, and the real exchange rate.

- Nominal GDP, exports, and nonoil imports are used to construct net exports relative to GDP.
- The terms of trade is measured as the price of nonoil imports relative to the price of exports.
- The real exchange rate is the Federal Reserve's trade-weighted dollar index.
- Series are Hodrick-Prescott filtered before computing business-cycle moments.

Interpretation: The macro comparison asks whether the sunk-cost model improves the standard international business cycle model along volatility, persistence, and comovement dimensions.

3.3 Exporter-count data

- Exporter data come from the U.S. Census Bureau's annual *Profile of U.S. Exporting Companies*.
- The sample covers the total number of U.S. exporters by destination market from 1995 to 2003.
- Destination countries include Australia, Belgium, Canada, France, Germany, Italy, Japan, Korea, Mexico, the Netherlands, Spain, Switzerland, and the United Kingdom.
- The paper links exporter counts to trade flows, U.S. GDP, destination-country GDP, and bilateral real exchange rates.

Interpretation: The ideal data would track firm-level exports by destination and source country. The available exporter-count data are less detailed, but still useful for checking whether the model's exporter dynamics are realistic.

3.4 Price-index and variety measurement

- The model contains an extensive margin: the number of imported varieties changes over time as firms enter and exit export markets.
- Welfare-based price indices can change when the number of varieties changes.
- Statistical price indices typically track average prices of continuing goods and may not fully incorporate the welfare value of additional varieties.
- The paper therefore studies whether a measured CPI-style price index materially changes real exchange rate moments.

Interpretation: The measurement issue matters because export entry changes not only quantities but also the set of goods available to consumers. In the benchmark model, however, the measured-versus-welfare price-index distinction has only small aggregate effects.

4 Models and empirical specifications

4.1 Consumer problem

The representative consumer in each country chooses consumption, labor, and state-contingent bond holdings. The benchmark period utility is

$$U(C,L) = \frac{\left[C^{\gamma}(1-L)^{1-\gamma}\right]^{1-\sigma}}{1-\sigma}.$$

- β is the discount factor.
- σ governs intertemporal substitution and risk aversion.
- γ governs the consumption weight in the consumption-leisure composite.
- Complete state-contingent bonds imply that the real exchange rate is proportional to the ratio of marginal utilities of consumption across countries.

Interpretation: The demand side is close to a standard international business cycle model. The paper's new economics mainly enters through production, export participation, and varieties.

4.2 Final goods and imported varieties

Final goods are produced competitively from domestic and foreign intermediate varieties. A stylized version of the aggregator is

$$D_t = \left\{ a_1 \left[\int_0^1 y_h(i)^\theta di \right]^{\rho/\theta} + (1 - a_1) [N_t^*]^{-\lambda} \left[\int_{\mathcal{E}_t^*} y_f(i)^\theta di \right]^{\rho/\theta} \right\}^{1/\rho}.$$

- a_1 controls home bias.
- $1/(1-\theta)$ is the elasticity of substitution across varieties produced in the same country.
- $1/(1-\rho)$ is the elasticity of substitution between home and foreign composite goods.
- N_t^* is the measure of foreign exporters selling in the home country.
- λ governs how the number of varieties affects utility and measured prices.

Interpretation: The model has both an intensive margin and an extensive margin. The central quantitative question is whether changing the extensive margin has enough bite to change aggregate dynamics.

4.3 International trade costs

- A firm can sell costlessly at home.
- To enter the foreign market, a nonexporter pays a high fixed cost τ_0.
- To continue exporting, a current exporter pays a lower fixed continuation cost τ_1.
- The benchmark has $\tau_0 > \tau_1$, which creates an inaction band and exporter hysteresis.
- These costs are denominated in units of labor in the destination market.

Interpretation: Export status is persistent because continuing exporters face a lower hurdle than new exporters. This generates a wedge between the productivity threshold for continuing and the threshold for entering.

4.4 Intermediate producers and productivity

Each intermediate producer makes a differentiated good using capital and labor:

$$y_{it} = A_{it} k_{it}^{\alpha} \ell_{it}^{1-\alpha}.$$

Productivity has an aggregate and firm-specific component:

$$\log A_{it} = z_t + \eta_{it}.$$

- Aggregate productivity follows a country-level vector autoregression.
- Firm-specific productivity is idiosyncratic and independent across firms.
- Firms differ in current productivity, capital, and export status.
- Firms choose prices, inputs, investment, and whether to export.

Interpretation: Heterogeneous productivity creates selection into exporting. Higher-productivity firms are more likely to find export-market participation worthwhile.

4.5 Export decision and thresholds

Let $V^1(\eta, k, m, s^t)$ be the value of exporting and $V^0(\eta, k, m, s^t)$ the value of not exporting, where m is lagged export status. The critical productivity thresholds solve

$$V^1(\eta_1, k, 1, s^t) = V^0(\eta_1, k, 1, s^t),$$

for continuing exporters and

$$V^1(\eta_0, k, 0, s^t) = V^0(\eta_0, k, 0, s^t),$$

for nonexporters.

Because $\tau_0 > \tau_1$, the entry threshold exceeds the continuation threshold:

$$\eta_0(s^t) > \eta_1(s^t).$$

The share of exporters evolves as

$$N_t = [1 - n_{1t}]N_{t-1} + n_{0t}[1 - N_{t-1}],$$

where n_{1t} is the exit probability among last-period exporters and n_{0t} is the entry probability among last-period nonexporters.

Interpretation: This is the model's hysteresis mechanism. Some current exporters keep exporting even though otherwise similar nonexporters would not enter.

4.6 Quantitative comparisons

The paper compares several economies:

- **No costs:** all firms can export without paying fixed costs.
- **Sunk cost benchmark:** entry and continuation costs create endogenous entry and exit.
- **Fixed exit:** exporters occasionally receive an exogenous continuation shock that makes them repay the entry cost.
- **Sensitivity cases:** high exporter persistence, i.i.d. exporters, love or hate of variety, higher markups, alternative Armington elasticity, and alternative measurement of price indices.

Interpretation: The comparison is designed to isolate whether the fixed-cost participation mechanism, not just firm heterogeneity, changes aggregate trade dynamics.

5 Identification and quantitative design

5.1 What the design can identify

- The paper is a quantitative theory paper, not a reduced-form causal design.
- It identifies whether a calibrated general equilibrium model with realistic exporter dynamics produces aggregate net export dynamics different from the standard model.
- The core claim is conditional: given the calibration and preferences, sunk costs alone have small aggregate effects.

Interpretation: The evidence is strongest as a quantitative mechanism test. It asks whether a plausible mechanism is big enough, not whether a policy shock causally changes exports.

5.2 Key calibration discipline

- The model matches basic U.S. macro moments and exporter facts.
- The benchmark cost parameters are $\tau_0 = 0.048$ and $\tau_1 = 0.010$.
- The aggregate productivity process has persistence around 0.95.
- The benchmark elasticity across foreign and home composites is lower than the within-country variety elasticity.

Interpretation: The small aggregate effect comes from the interaction of trade costs, product substitution, and variety valuation. The conclusion should not be read as saying export frictions never matter in any model.

5.3 Selection and aggregation

- Export costs strongly affect which firms export.
- But aggregate imports and exports depend mainly on total foreign-good demand, not on exactly how that demand is distributed across exporters.
- When varieties within a country are close substitutes, the number of exporters is not a powerful aggregate state variable.

Interpretation: The paper's most important aggregation lesson is that large micro reallocation does not necessarily create large macro movement.

5.4 Main limitations

- The model abstracts from firm births and deaths, focusing only on export entry and exit among existing firms.
- Idiosyncratic productivity shocks are simplified to keep the state space manageable.

- The model has standard international business cycle shortcomings, including too little cross-country comovement and too little real exchange rate volatility.
- The exporter data are aggregate counts by destination, not a complete firm-by-destination panel.

Interpretation: The paper's negative result is powerful, but it is not the final word on all trade frictions. The authors explicitly suggest that other frictions may be more important for net export dynamics.

6 Tables and figures

6.1 Figure I and Table I: Export hysteresis mechanism and calibration

Read:

- Figure I shows firm value as a function of idiosyncratic productivity.
- Without export costs, exporting dominates because it gives access to a larger market.
- With trade costs, the value of exporting shifts down. The shift is larger for new exporters because they pay τ_0 instead of τ_1.
- The continuation threshold η_1 is lower than the entry threshold η_0, creating exporter hysteresis.
- Table I lists the benchmark parameters and the alternative parameterizations used in sensitivity analysis.
- The benchmark uses $\beta = 0.99$, $\sigma = 2$, $\theta = 0.9$, $\rho = 1/3$, $\alpha = 0.36$, $\delta = 0.025$, $\tau_0 = 0.048$, and $\tau_1 = 0.010$.

Economic message: Figure I is the micro foundation of the paper: sunk entry costs create different entry and continuation cutoffs. Table I then asks whether this mechanism is quantitatively important once embedded in a full business cycle model.

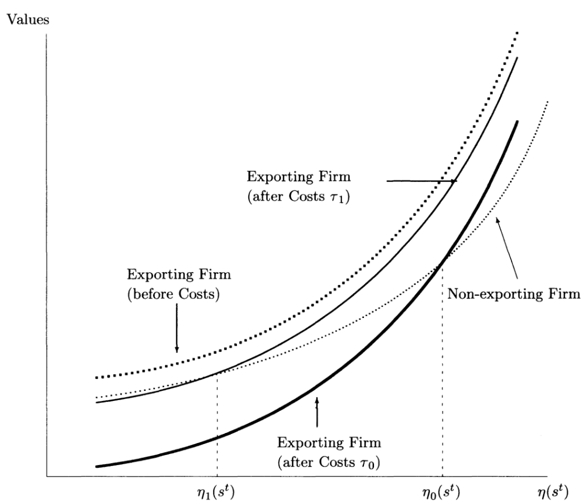


FIGURE I
Value of Firms

(a) Figure I: Value of firms and export thresholds.

TABLE I
PARAMETER VALUES

Benchmark Model	
Preferences	$\beta = 0.99$, $\sigma = 2$, $\theta = 0.9$, $\rho = 1/3$, $\gamma = 0.303$
Production	$\alpha = 0.36$, $\delta = 0.025$, $a_1 = 0.757$, $\lambda = 0$
Productivity	$M_{11} = M_{22} = 0.95$, $M_{12} = M_{12} = 0$,
	Var (ϵ) = Var $(\epsilon^*) = \sigma_\epsilon^2 = 0.007^2$
	Corr $(\epsilon, \epsilon^*) = 0.25$, $\sigma_\eta = 0.50$
Trade costs	$\tau_0 = 0.048$, $\tau_1 = 0.010$
Variations	
No cost	$\gamma = 0.308$, $a_1 = 0.761$, $\tau_0 = \tau_1 = 0$
Fixed exit	$a_1 = 0.757$, $\tau_0 = 0.168$, $\tau_1 = 0$, Pr (exit shock) = 0.047
Permanent exporters	$a_1 = 0.756$, $\tau_0 = 0.181$, $\tau_1 = 0.012$
IID exporters	$\gamma = 0.306$, $a_1 = 0.759$, $\tau_0 = \tau_1 = 0.005$
Love variety	$\lambda = -0.1$, $a_1 = 0.752$, $\tau_0 = 0.048$, $\tau_1 = 0.01$
Hate variety	$\lambda = 1$, $a_1 = 0.801$, $\tau_0 = 0.048$, $\tau_1 = 0.01$
High markup	$\theta = 2/3$
No cost	$\gamma = 0.393$, $a_1 = 0.761$
Sunk	$\gamma = 0.368$, $a_1 = 0.742$, $\tau_0 = 0.144$, $\tau_1 = 0.047$
CRS	$\lambda = 1/3$, $\gamma = 0.368$, $a_1 = 0.763$, $\tau_0 = 0.144$, $\tau_1 = 0.047$
High armington	$\rho = \theta = 2/3$
No cost	$\alpha_1 = 0.641$
Sunk-CRS	$\lambda = 1/3$, $a_1 = 0.647$, $\tau_0 = 0.16$, $\tau_1 = 0.052$
Annual	$M_{11} = M_{22} = 0.95^4$, $\sigma_\epsilon^2 = 0.0242^2$, $\sigma_\eta = 0.25$,
	$\beta = 0.96$, $\gamma = 0.304$, $\delta = 0.10$, $\tau_0 = 0.021$, $\tau_1 = 0.01$

(b) Table I: Parameter values.

6.2 Figure II and Figure III: Exporter characteristics and aggregate impulse responses

Read:

- Figure II varies the probability of exiting the export market and reports exporter premia.
- When exit is unlikely, exporters have a larger capital premium because they expect to keep serving the larger foreign market.
- When export participation is closer to i.i.d., exporter productivity, output, and labor premia are higher, but the capital premium is lower.
- Figure III compares impulse responses after a one-standard-deviation home productivity shock.
- The models generate different exporter participation dynamics, but net exports and real exchange rates look almost identical across the no-cost, sunk-cost, and fixed-exit economies.

Economic message: Exporter hysteresis matters for who exports and for exporter characteristics, but it barely changes the aggregate response of the trade balance or real exchange rate in the benchmark economy.

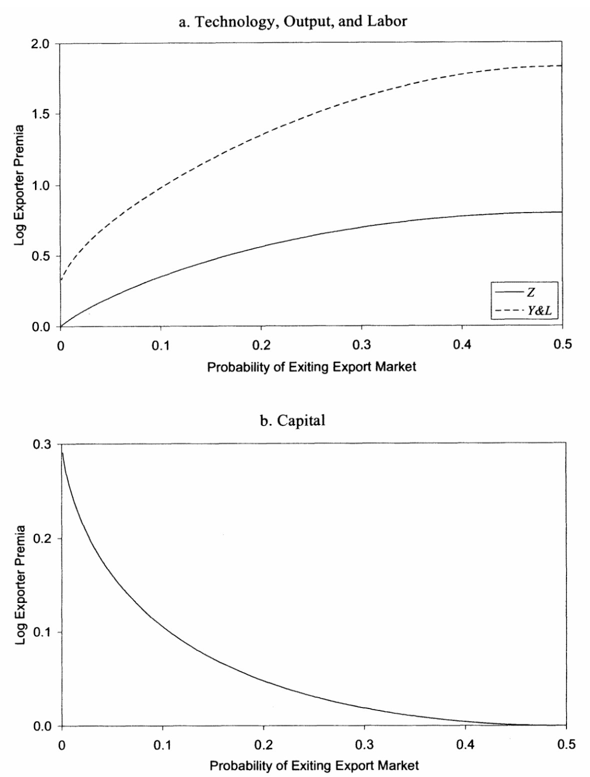


FIGURE II
Exporter Characteristics and Hysteresis

(a) Figure II: Exporter characteristics and hysteresis.

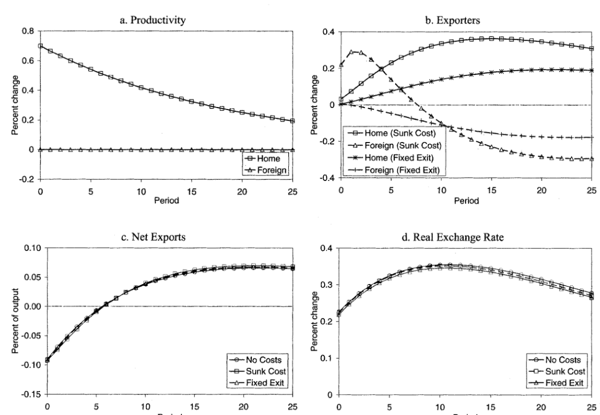


FIGURE III
Response to a Home Aggregate Productivity Shock

(b) Figure III: Response to a home aggregate productivity shock.

6.3 Figure IV and Figure V: Sequential shocks and cross-correlations

Read:

- Figure IV studies a home productivity shock followed later by a foreign productivity shock.
- The second shock creates sharp changes in export participation, but the real exchange rate and net export paths remain nearly identical in the no-cost and sunk-cost models.
- Figure V plots the cross-correlation between net exports and the real exchange rate.
- The no-cost, sunk-cost, and fixed-exit models almost sit on top of one another.
- The result suggests that the fixed cost of exporting does not shift the timing of the aggregate net export-real exchange rate relationship.

Economic message: The key macro result survives a dynamic shock sequence. Even when exporter counts move differently, the aggregate cross-correlation pattern barely changes.

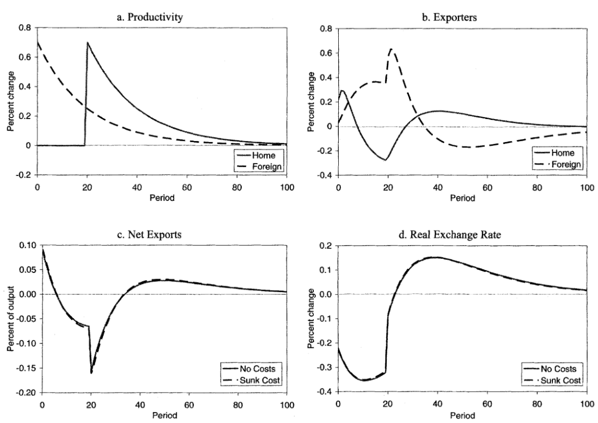


FIGURE IV
Response to a Home Productivity Shock Followed by a Foreign Productivity Shock

(a) Figure IV: Home shock followed by foreign shock.

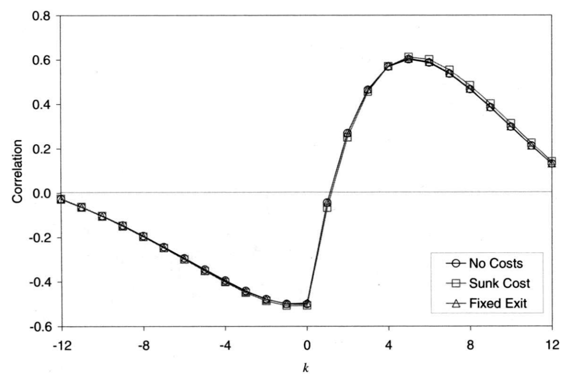


FIGURE V
Correlation of nx_{t+k} and q_t

(b) Figure V: Correlation of nx_{t+k} and q_t.

6.4 Table II: Business cycle statistics

Read:

- Table II compares data, the no-cost model, the sunk-cost model, fixed-exit alternatives, and sensitivity cases.
- The no-cost and benchmark sunk-cost columns are very similar for output volatility, net export volatility, relative standard deviations, domestic correlations, persistence, and international correlations.
- Sunk costs slightly increase net export persistence and slightly reduce international comovement, but the magnitudes are small.
- The most important sensitivity cases are the love-of-variety and hate-of-variety columns. These change international risk sharing and business-cycle synchronization more than the sunk-cost friction itself.

- The high-Armington case makes home and foreign goods more substitutable and worsens the model's fit to business-cycle comovement and real exchange rate behavior.
- The measured-price-index column produces only minor differences from the benchmark in this calibration. The table is split into two direct extracts below to preserve portrait orientation and readability.

Economic message: Table II is the core quantitative table. It shows that the macro economy is nearly invariant to the addition of sunk exporting costs, while the valuation of variety and substitution elasticities matter much more.

	Data	No costs	Sunk cost	Fixed exit	High persistence
Standard deviation (in percent)					
Y	1.42	1.27	1.28	1.28	1.29
nx	0.46	0.16	0.16	0.16	0.16
Standard deviation (relative to output)					
C	0.79	0.36	0.36	0.36	0.36
I	3.25	3.36	3.37	3.36	3.36
L	0.85	0.47	0.47	0.47	0.46
q	2.81	0.33	0.32	0.32	0.32
Domestic correlation with output					
C	0.83	0.95	0.95	0.95	0.95
I	0.93	0.98	0.98	0.98	0.98
L	0.85	0.99	0.99	0.99	0.99
nx	−0.38	−0.49	−0.50	−0.50	−0.50
q	0.16	0.56	0.56	0.56	0.57
q, nx	0.07	−0.50	−0.51	−0.50	−0.51
Persistence					
Y	0.87	0.69	0.69	0.69	0.69
C	0.91	0.73	0.73	0.73	0.74
I	0.84	0.67	0.68	0.67	0.68
L	0.95	0.68	0.68	0.68	0.68
nx	0.90	0.69	0.71	0.70	0.72
q	0.81	0.79	0.79	0.79	0.79
International correlation[a]					
Y	0.51	0.22	0.22	0.21	0.21
C	0.32	0.54	0.54	0.55	0.54
I	0.29	0.02	0.00	0.00	0.01
L	0.43	0.19	0.19	0.18	0.19

[a] International correlations are from Kehoe and Perri [2000].

(a) Table II, part 1: Data, no-cost, sunk-cost, fixed-exit, and high-persistence columns.

	IID exporters	Love variety	Hate variety
Standard c			
Y	1.28	1.30	1.19
nx	0.16	0.19	0.08
Standard c			
C	0.36	0.36	0.37
I	3.36	3.43	3.18
L	0.47	0.47	0.46
q	0.33	0.32	0.38
Domestic c			
C	0.95	0.95	0.96
I	0.98	0.97	0.99
L	0.99	0.98	0.99
nx	−0.49	−0.50	−0.45
q	0.56	0.57	0.55
q, nx	−0.50	−0.51	−0.49
Persistenc			
Y	0.69	0.69	0.68
C	0.73	0.74	0.73
I	0.67	0.69	0.66
L	0.68	0.69	0.66
nx	0.69	0.73	0.63
q	0.79	0.80	0.77
Internatio			
Y	0.22	0.21	0.29
C	0.54	0.55	0.48
I	0.01	−0.04	0.17
L	0.19	0.17	0.28

[a] Intern

(b) Table II, part 2: IID-exporter and variety columns.

	High markup		
	No costs	Sunk cost	CRS
Standard c			
Y	1.22	1.29	1.22
nx	0.10	0.15	0.09
Standard c			
C	0.41	0.40	0.41
I	3.91	4.07	3.87
L	0.42	0.42	0.40
q	0.41	0.38	0.42
Domestic c			
C	0.95	0.95	0.96
I	0.98	0.96	0.98
L	0.98	0.98	0.98
nx	−0.36	−0.41	−0.35
q	0.59	0.61	0.59
q, nx	−0.32	−0.40	−0.31
Persistenc			
Y	0.69	0.70	0.69
C	0.73	0.73	0.73
I	0.67	0.70	0.68
L	0.68	0.70	0.68
nx	0.75	0.81	0.78
q	0.77	0.78	0.77
Internatio			
Y	0.21	0.17	0.21
C	0.60	0.61	0.58
I	0.04	−0.07	0.07
L	0.13	0.11	0.19

[a] Intern

(c) Table II, part 3: High-markup columns.

		ρ = θ		
		No costs	Sunk-CRS	Measure
Standard c				
Y	2	1.30	1.29	1.28
nx	9	0.20	0.17	0.16
Standard c				
C	1	0.37	0.38	0.35
I	7	4.32	4.16	3.36
L	0	0.49	0.45	0.47
q	2	0.24	0.27	0.32
Domestic c				
C	6	0.90	0.91	0.95
I	8	0.95	0.96	0.98
L	8	0.98	0.98	0.99
nx	5	−0.07	0.02	−0.50
q	9	0.59	0.60	0.55
q, nx	1	0.38	0.46	−0.46
Persistenc				
Y	9	0.70	0.70	0.69
C	3	0.73	0.73	0.73
I	8	0.67	0.68	0.68
L	8	0.68	0.68	0.68
nx	8	0.87	0.93	0.71
q	7	0.85	0.83	0.80
Internatio				
Y	1	0.07	0.08	0.22
C	8	0.74	0.71	0.53
I	7	−0.24	−0.17	0.00
L	9	−0.24	−0.12	0.19

[a] Intern

(d) Table II, part 4: High-Armington and measured-price columns.

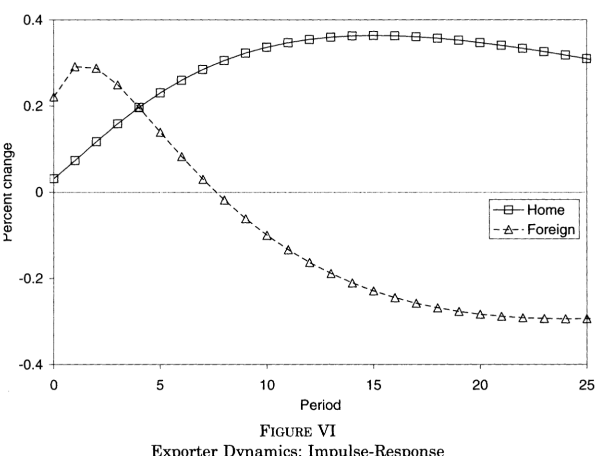
6.5 Figure VI and Figure VII: Exporter dynamics in the model and data

Read:

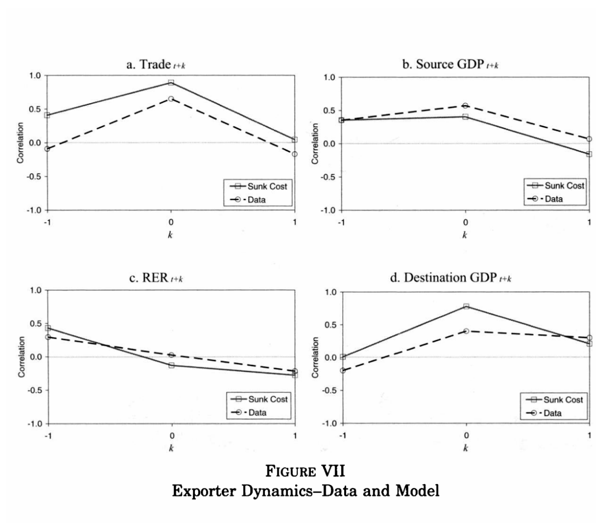
- Figure VI shows exporter-count dynamics after a home productivity shock.
- Home exporters expand gradually and persistently because home firms receive a sustained cost advantage in foreign markets.
- Foreign exporters initially expand because the home boom raises demand for foreign goods, but then contract as the relative cost advantage shifts toward home firms.
- Figure VII compares model and data moments for the number of U.S. exporters by destination market.
- Exporter counts are strongly related to trade flows and lag source-country GDP.

- The model also reproduces the limited contemporaneous relationship between exporters and the real exchange rate.
- The weakest fit is destination GDP: in the data, U.S. exporters appear to enter in anticipation of foreign expansions, while the model predicts more contemporaneous comovement.

Economic message: The model is much better at matching the timing of exporter participation than at changing aggregate net export dynamics. This reinforces the paper's distinction between firm-level export decisions and aggregate trade-balance dynamics.



(a) Figure VI: Exporter dynamics impulse response.



(b) Figure VII: Exporter dynamics in data and model.

7 Short summary

- The paper takes a well-known micro friction - sunk export-market entry costs - and embeds it in a general equilibrium international business cycle model.
- The model generates exporter hysteresis: continuing exporters have a lower productivity threshold than nonexporters considering entry.
- The model also generates realistic exporter premia and exporter-count dynamics.
- However, the aggregate net export and real exchange rate dynamics are nearly identical to those in a model without sunk export costs.
- The reason is aggregation: if goods from the same country are close substitutes, the number of exporters has little independent effect on total trade quantities.
- The main way export participation can matter for aggregate dynamics is through the number of varieties and the value consumers attach to variety.
- The authors conclude that lags in expanding trade flows may be more important for net export dynamics than the fixed costs of entering or continuing to export.

8 Conclusion

8.1 Core conclusion

Sunk costs of exporting matter for individual firms, but they do not materially alter aggregate net export or real exchange rate dynamics in the benchmark international business cycle model. The standard model and the exporter-hysteresis model behave surprisingly similarly at the aggregate level.

8.2 Economic intuition

The micro friction changes participation, but aggregate trade depends mostly on total demand for foreign composites. When varieties within a country are close substitutes, changing the number of exporters does not strongly change the quantity of imported goods. The economy reallocates trade across exporters rather than changing aggregate trade dynamics.

8.3 Incremental contribution revisited

The paper's main contribution is not just adding sunk costs to a model. It demonstrates that a plausible micro friction, even one supported by exporter data, may fail to generate the macro mechanism often attributed to it. This is an important discipline for theories of the J-curve and export hysteresis.

8.4 Implications for trade and macro models

Models that seek to explain delayed net export responses may need to focus on additional margins: time-to-ship, time-to-build export capacity, customer accumulation, pricing frictions, search and distribution networks, or other mechanisms that delay quantities directly. Fixed costs of participation alone are not enough in the benchmark calibration.

8.5 Implications for welfare and variety

Because the main aggregate effect of export participation runs through the variety margin, welfare conclusions depend heavily on the taste for variety. If consumers value additional varieties strongly, exporter entry can matter more. If the gains to variety are small or negative, the welfare benefits of trade liberalization through variety expansion may be overstated.

8.6 Limitations

The model abstracts from firm births and deaths, richer firm-level export growth patterns, and many macro frictions. It also inherits standard international business cycle problems, including too-smooth real exchange rates and insufficient cross-country comovement. The exporter-count data are useful but not as rich as a firm-by-destination panel.

8.7 Takeaway

The paper's central lesson is an aggregation lesson: do not infer large macro trade effects from the existence of micro export hysteresis. The extensive margin of exporting is important for firms, but its macro importance depends on substitution patterns, variety valuation, and the presence of other frictions that directly slow trade quantities.